Replicating “End-to-End Text to Speech Synthesis”: Results Summary

CS 229 replication project | [Team names] | [Date]

Overview

We replicated the experiment of Wang, Yang and Li (CS 229, Autumn 2018), which compares three models for generating speech directly from text: support vector regression (SVR), a simple neural network (NN), and a Seq2Seq model with attention. We trained on LJ Speech with 200 training sentences (plus 20 dev sentences), using only clips of 5 seconds or less. The pipeline is: text, CMU phoneme/letter mapping, model, spectrogram, inverse STFT and inverse FIR filter, wav. Quality was rated by listeners with the Mean Opinion Score (MOS, 1 to 5).

Results

Model	Paper MOS	Our MOS	Std	Listeners
SVR	1.0	1.0	0.00	5
NN	1.7	1.4	0.49	5
Seq2Seq	2.5	1.6	0.49	5

Table 1. MOS from our listeners next to the paper's values. Std is the population standard deviation.

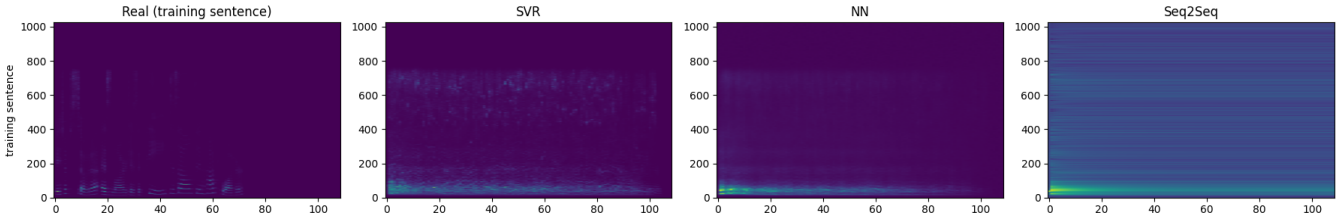


Figure 1. Spectrograms of a real training sentence and the SVR, NN and Seq2Seq predictions for it.

What the audio sounds like:

- **SVR** is the worst of the three.
- **NN** is louder than SVR but still poor.
- **Seq2Seq** is equally poor but has a repeating rhythm (beats).
- The **zero-phase ground truth** sounds buzzy, which sets the ceiling for this pipeline.

Why the models behave this way

SVR outputs near-constant values because its default epsilon (0.1) is larger than almost all spectrogram amplitudes (mean 0.00026, max 0.199). The NN smears its output toward an average spectrum. Seq2Seq feeds its own output back at every step and falls into a loop, which produces the repeating beat. Because only magnitude is predicted, phase is lost and every output sounds robotic. Training loss is also misleading: amplitudes are tiny, so all losses look small, but dev MSE never beats predicting all zeros (2.37×10^{-6}): SVR 3.06×10^{-5}, NN 8.57×10^{-6}, Seq2Seq (free-running) 2.42×10^{-6}.

Comparison with the paper and limitations

The ordering of the models matches the paper (SVR < NN < Seq2Seq), but our numbers are not identical and the gaps are smaller (1.0 / 1.4 / 1.6 versus 1.0 / 1.7 / 2.5). With only 5 listeners, we do not claim the differences are significant. Limitations: clips are capped at 5 s (the paper has no cap); many hyperparameters are our guesses (FIR filter with 101 taps and an 8 kHz cutoff, STFT hop of 50%, phoneme probability 0.5, embedding sizes, Adam with MSE, scikit-learn SVR defaults); and 200 training sentences is a very small set.

Conclusion

Seq2Seq with attention scored highest, as in the paper, but none of the three models produces natural speech. Zero phase and a small training set limit all of them; a deeper model, more data and a phase-recovery step such as Griffin-Lim are the obvious next steps.

Reference: X. Wang, Y. Yang and Y. Li, “End-to-End Text to Speech Synthesis,” CS 229 Project Report, Stanford University, Autumn 2018.